

An n^2 -vertex characterization of inverse-M interval matrices

Repository IV-03. Claimed resolution; independent review pending.

Abstract

An interval matrix consists entirely of inverse-M matrices if and only if n^2 specified vertices are inverse-M matrices. This proves the proposed $2n^2$ -vertex criterion and shows that half of those tests are unnecessary. The proof combines induction on principal submatrices, monotonicity of two-index Schur complements, and an adjugate lemma that rules out singularity without assuming regularity in advance. Zero entries, zero interval widths, and reducible matrices are allowed.

1 Definitions and main theorem

A nonsingular M-matrix is a real matrix B with nonpositive off-diagonal entries and $B^{-1} \geq 0$. A matrix A is inverse-M if it is nonsingular, $A \geq 0$, and A^{-1} has nonpositive off-diagonal entries. These definitions include reducible matrices.

Let $\mathbf{A} = [L, U] = [C - R, C + R]$, where $R \geq 0$. For $i = 1, \dots, n$, let $z^{(i)}$ have entry -1 at position i and entry 1 elsewhere, and put $D_i = \text{diag}(z^{(i)})$. Define

$$V_{ij} = C - D_i R D_j \quad (1 \leq i, j \leq n).$$

The repository's proposed polynomial criterion uses both signs of this expression [1, 2]; an exponential vertex characterization is known [3].

Theorem 1. *Every matrix in $\mathbf{A}$ is inverse-M if and only if every one of the n^2 matrices V_{ij} is inverse-M.*

The reverse implication is the substantive one. Since these n^2 vertices are among the proposed $2n^2$ tests, the theorem implies that proposed characterization as well.

2 Basic closure facts

Lemma 2. *Every principal submatrix of a nonsingular M-matrix is a nonsingular M-matrix, and all its principal minors are positive. Every principal submatrix of an inverse-M matrix is inverse-M and has positive determinant. Every principal Schur complement of an inverse-M matrix is inverse-M, in particular entrywise nonnegative.*

Proof. Let B be a nonsingular M-matrix. Set $w = B^{-1}\mathbf{1} > 0$; strict positivity follows because a nonnegative invertible matrix has no zero row. The matrix $H = B \text{diag}(w)$ has nonpositive off-diagonal entries and satisfies $H\mathbf{1} = \mathbf{1}$. Thus H and all its principal submatrices are strictly row diagonally dominant with positive diagonal. Write any such submatrix as $D(I - K)$ with positive diagonal D , $K \geq 0$, and $\|K\|_\infty < 1$. The Neumann series gives a nonnegative inverse. The determinant is positive because $\det(I - tK)$ is continuous and nonzero for $0 \leq t \leq 1$, and is 1 at $t = 0$. Positive diagonal scaling transfers these conclusions to the principal submatrices of B .

Now let $A = B^{-1}$ and partition the indices as S, T . Jacobi's complementary principal-minor identity gives

$$\det A_{SS} = \frac{\det B_{TT}}{\det B} > 0,$$

with the empty determinant interpreted as 1. Moreover,

$$A_{SS}^{-1} = B_{SS} - B_{ST}B_{TT}^{-1}B_{TS}$$

has nonpositive off-diagonal entries. Since $A_{SS} \geq 0$, it is inverse-M. Finally,

$$A_{TT} - A_{TS}A_{SS}^{-1}A_{ST} = (B_{TT})^{-1}$$

is inverse-M and nonnegative. $\square$

Lemma 3 (Adjugate completion). *Let $n \geq 3$. If all nonempty proper principal minors of A are positive and all off-diagonal entries of $\text{adj } A$ are nonpositive, then $\det A > 0$.*

Proof. Suppose first $\det A = 0$. A nonzero principal minor of order $n - 1$ implies $\text{rank } A = n - 1$. Thus $H = \text{adj } A$ has rank one and strictly positive diagonal. Rank one implies

$$H_{12}H_{23}H_{31} = H_{11}H_{22}H_{33} > 0,$$

contradicting nonpositivity of the three off-diagonal factors.

Suppose instead $\det A < 0$, and put $B = A^{-1}$. Its diagonal is negative and its off-diagonal entries are nonnegative. Jacobi's identity implies that every nonempty principal minor of B is negative, since its complementary principal minor in A is positive, or equals 1 when the complement is empty. Choose any three indices and write the corresponding principal submatrix as

$$\begin{pmatrix} -a & p & q \\ r & -b & s \\ t & u & -c \end{pmatrix}, \quad a, b, c > 0, \quad p, q, r, s, t, u \geq 0.$$

Its negative principal minors of order two imply $pr > ab$, $qt > ac$, and $su > bc$. Its determinant is

$$-abc + asu + bqt + cpr + pst + qru > 2abc > 0,$$

a contradiction. Both alternatives are impossible. $\square$

3 Proof of the vertex characterization

Assume all V_{ij} are inverse-M. Each lower endpoint is nonnegative, since $(V_{ij})_{ij} = L_{ij}$. Thus $L \geq 0$ and every A in the interval is nonnegative.

We induct on n . For $n = 1$, the tested matrix is L , and positivity of L suffices. For $n = 2$, the vertex

$$V_{11} = \begin{pmatrix} L_{11} & U_{12} \\ U_{21} & L_{22} \end{pmatrix}$$

has positive determinant. Hence every interval member has

$$\det A = a_{11}a_{22} - a_{12}a_{21} \geq L_{11}L_{22} - U_{12}U_{21} > 0.$$

Its inverse has nonpositive off-diagonal entries, proving the assertion.

Let $n \geq 3$. Every proper principal interval inherits the vertex hypothesis: the local test for indices i, j is the corresponding principal submatrix of the global V_{ij} . Lemma 2 and induction imply that every proper principal submatrix of every $A \in \mathbf{A}$ is inverse-M. In particular all proper principal minors of every interval member are positive. This is an inductive proof, not an instruction to perform exponentially many algorithmic tests.

Fix distinct i, j , and put $S = \{1, \dots, n\} \setminus \{i, j\}$. Define

$$f_{ij}(A) = a_{ij} - A_{iS}A_{SS}^{-1}A_{Sj}, \quad u = A_{iS}A_{SS}^{-1}, \quad v = A_{SS}^{-1}A_{Sj}.$$

Here $u \geq 0$ and $v \geq 0$. For example, in the proper principal inverse-M block on $\{i\} \cup S$, the off-diagonal block of its inverse is $-\alpha^{-1}u$, where the scalar Schur complement $\alpha > 0$ by the ratio of positive principal determinants. Its nonpositivity proves $u \geq 0$. The argument for v uses the block on $S \cup \{j\}$.

Throughout the box, ordinary matrix differentiation gives

$$\frac{\partial f_{ij}}{\partial a_{ij}} = 1, \quad \frac{\partial f_{ij}}{\partial a_{ik}} = -v_k, \quad \frac{\partial f_{ij}}{\partial a_{kj}} = -u_k, \quad \frac{\partial f_{ij}}{\partial a_{k\ell}} = u_k v_\ell \quad (k, \ell \in S).$$

All other entries are irrelevant to f_{ij} . These signs show that its minimum is attained with a_{ij} and A_{SS} at their lower endpoints, and A_{iS} and A_{Sj} at their upper endpoints. These are exactly their values in V_{ij} . Consequently

$$f_{ij}(A) \geq f_{ij}(V_{ij}) \geq 0.$$

The last inequality holds because $f_{ij}(V_{ij})$ is an off-diagonal entry of the two-index principal Schur complement of an inverse-M matrix; Lemma 2 makes that complement nonnegative.

The adjugate identity

$$(\text{adj } A)_{ij} = -\det(A_{SS})f_{ij}(A)$$

now proves that every off-diagonal adjugate entry is nonpositive. This identity does not require A itself to be invertible: it follows by eliminating the invertible block A_{SS} and taking the appropriate cofactor of the resulting two-by-two block, or by polynomial continuity. Lemma 3 yields $\det A > 0$. Hence A^{-1} has nonpositive off-diagonal entries. Since $A \geq 0$, it is inverse-M. The forward implication of Theorem 1 is immediate because the test matrices belong to the interval.

4 Complexity and boundary cases

Form n^2 rational vertices. For each, check nonsingularity and entrywise nonnegativity, compute its inverse exactly, and check its off-diagonal signs. This takes $O(n^5)$ arithmetic operations using standard cubic inversion, with polynomial rational bit complexity. The arithmetic-operation count is not a claim of strongly polynomial bit complexity.

No strict positivity of off-diagonal entries or interval widths was assumed. The proof explicitly treats possible singularity of an interior full matrix through Lemma 3, rather than presuming regularity to justify inversion. Reducible and point intervals are included.

References

- [1] OpenProblemsInNLA, problem IV-03. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-03>.
- [2] M. Hladík, *An overview of polynomially computable characteristics of special interval matrices*, Springer (2020), 295–310, Section 9. https://doi.org/10.1007/978-3-030-31041-7_16. Author version: <https://arxiv.org/abs/1711.08732>.
- [3] J. Garloff, A. Al-Saafin and M. Adm, *Further Matrix Classes Possessing the Interval Property*, Reliable Computing 28 (2021), 56–70, Section 4. <https://reliable-computing.org/reliable-computing-28-pp-056-070.pdf>.